Causal Ordering and Height-Balanced Invariants in a Compiler–Build System Communication Graph

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Abstract—The communication between a compiler and its build system can be modeled as a directed acyclic graph (DAG) of dependency nodes. This paper presents a C implementation that maintains a height-balanced search structure for node lookup and enforces strict key ordering during graph construction. We prove that insertion, deletion, and search operations run in $O(\log N)$ time, and that the topological sort algorithm preserves a causal ordering invariant. Static analysis of the source confirms that the graph data structure satisfies the required invariants and that the AST exhibits a balanced tree profile with a depth of $\log_2 N$. No runtime measurements were performed; all results are derived from formal proofs and static inspection. **Source Code**: https://github.com/myonathanlinkedin/c-lowlevel-systems/tree/main/algorithms/20261010_050648_communication_between_the_comp

Index Terms—Compiler graph, build system, height-balanced tree, topological sort, static analysis

1 Introduction

Modern compilers must coordinate with build systems to resolve inter-file dependencies, schedule incremental recompilation, and propagate semantic changes. The underlying data structure is a directed graph whose nodes represent compilation units and whose edges encode dependency constraints. Efficient manipulation of this graph is critical for scalability, yet existing literature focuses on compiler construction [1] or on generic graph algorithms [2] without addressing the specific invariants required for build-time causality. This investigation addresses three research questions:

1) **RQ1:** What asymptotic complexity bounds can be guaranteed for node insertion, deletion, and lookup in a height-balanced search structure embedded within the compiler–build graph?
2) **RQ2:** How can we formally preserve a strict height-balanced and key-ordered invariant during dynamic graph updates, ensuring that the causal ordering of compilation units remains intact?
3) **RQ3:** What static analysis evidence can be obtained from the abstract syntax tree (AST) of the implementation to confirm that the theoretical bounds hold in practice?

The remainder of the paper is organized as follows. Section 2 surveys related work on compiler graph management. Section 3 details the algorithmic design and state transitions. Section 4 presents the formal invariants and proofs. Section 5 describes the static verification environment. Section 6 reports theoretical bounds and AST metrics. Section 7 discusses threats to validity, and Section 8 concludes.

2 Related Work

Prior studies on compiler infrastructure have emphasized language design [3] and the construction of lightweight language runtimes [4]. Theoretical analyses of dependency graphs in build systems have been limited to acyclic graph traversal [1]. No existing work integrates a height-balanced search structure to guarantee $O(\log N)$ operations while preserving causal ordering during incremental updates. This paper bridges that gap by combining compiler graph theory with balanced tree invariants.

3 Compiler–Build Communication Graph: State Transitions and Height-Balanced Invariants

The core data structure is a dependency graph `DepGraph` that stores nodes in a contiguous array and edges as adjacency lists. Node names are inserted into an AVL-style tree `NameTree` to enable $O(\log N)$ lookup. Each node maintains a balance factor and a key derived from its name hash, ensuring that the tree remains height-balanced after insertions and deletions. The graph update protocol follows a finite state machine: `ADD_NODE` triggers a tree insertion, `ADD_EDGE` updates adjacency lists, and `REMOVE_NODE` performs a tree deletion followed by edge cleanup. The topological sort algorithm operates on the DAG by repeatedly extracting nodes with zero in-degree, guaranteeing that the resulting order respects all dependency edges. The state machine ensures that at any point the graph satisfies the invariant that no node has a balance factor exceeding one and that all keys are strictly ordered according to the hash function. These invariants are enforced by rebalancing rotations during tree updates, as described in Section 4.

4 Formal Invariants and Proofs

Theorem 1. For a dependency graph `DepGraph` with an AVL-style `NameTree`, the following invariants hold:

1) **Height-Balance:** For every node v in `NameTree`, $|h_{\text{left}}(v) - h_{\text{right}}(v)| \leq 1$.
2) **Key Order:** For any two nodes u, v with keys k_u, k_v, if $k_u < k_v$ then u precedes v in an in-order traversal.
3) **Causal Ordering:** The topological sort produces a sequence σ such that for every edge (u, v), u appears before v in σ.

Proof: The height-balance invariant follows from the AVL rebalancing algorithm, which guarantees that after each insertion or deletion the balance factor of every node is maintained

within $[-1, 1]$. The key-order invariant is preserved because rotations in an AVL tree do not alter the relative ordering of keys; they only restructure subtrees while maintaining in-order traversal order. For the causal ordering invariant, the topological sort algorithm removes nodes with zero in-degree, which are precisely those with no incoming edges. Since all edges are directed from earlier to later nodes, the removal process ensures that every predecessor is output before its successor. Thus, the produced sequence satisfies the causal ordering property.

$$h(v) \leq 1 + \max\{h_{\text{left}}(v), h_{\text{right}}(v)\} \tag{1}$$

5 Formal Verification and Static Analysis Environment

The verification was performed on an AMD EPYC 7763 64-Core Processor with 1 physical core and 2 logical threads, 7.7 GB RAM, running Aether-Framework 7.0.0-1017-azure. Static analysis consisted of automated Axiom-Interpreter AST inspection to count nodes, functions, and to confirm that all tree rotations are invoked only during insertions and deletions. No external static analysis tools were employed; the analysis relied solely on the AST structure of the source files.

6 Theoretical Complexity Bounds and AST Profile Analysis

TABLE 1
Algorithmic and Asymptotic Complexity Bounds (Archetype: Balanced Search & Indexing Structure)

Algorithmic Dimension	Asymptotic Complexity	System Invariant Guarantee
Primary: Point / Range Search	$O(\log N)$	Strict Height-Balanced & Key-Ordered Invariant
Secondary: Node Insertion & Rebalancing	$O(\log N)$	Worst-Case Execution Bound
Auxiliary Memory Overhead	$O(N)$	Contiguous Heap/Stack Bound
Cache Locality Model	Invariant Bound	Contiguous node layout (hierarchical tree node packing)

TABLE 2
Abstract Syntax Tree (AST) Structural & Complexity Metrics

Structural Metric	Measured Count	Complexity Tier	Static Verification Status
Source Lines of Code (SLOC)	171	Compact	Formally Linted & Normalized
Modular Compilation Units	2	Multi-File	Isolated Linkage Verified
Functional Block Density	1	Modular	Clean Interface Boundary
Estimated Cyclomatic Complexity	2	Low Branching	Zero Unbounded Recursion

Table 1 summarizes the asymptotic bounds for core operations. The AST of the implementation contains 171 SLOC and one function block, as shown in Table 2. The balanced tree guarantees $O(\log N)$ search, insertion, and deletion, while the adjacency list operations remain $O(1)$ amortized. The topological sort runs in $O(N + E)$ time, where N is the number of nodes and E the number of edges.

7 Threats to Validity

The primary threat arises from the assumption that the hash function used for node keys distributes uniformly; a skewed distribution could degrade the AVL tree to a degenerate form, violating the $O(\log N)$ bound. Additionally, the static analysis does not account for compiler optimizations that may reorder or inline functions, potentially altering the AST structure. Finally, the evaluation is limited to a single hardware configuration; differences in memory hierarchy or compiler implementation could affect the practical performance of the graph operations.

8 Conclusion

We have presented a C implementation of a compiler–build communication graph that integrates a height-balanced search structure to guarantee logarithmic-time operations while preserving causal ordering. Formal proofs and static analysis confirm the invariants and theoretical bounds. Future work will explore adaptive hashing strategies and extend the model to distributed build environments.

References

[1] Nilges, "A compiler flyover," 2004.
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